Lösungen · Trigonometrie

1. $\varepsilon = \cos^{-1}(\frac{13}{32}) \approx 66{,}0°$ ... $\cos\varepsilon = \frac{13}{32}$

2. $\beta_1 = \cos^{-1}(\frac{255}{384}) \approx 48{,}4°$ ... $\cos\beta_1 = \frac{255}{384}$

3. $\alpha = \sin^{-1}(\frac{7{,}4}{14{,}1}) \approx 31{,}7°$ ... $\sin\alpha = \frac{7{,}4}{14{,}1}$

4. $\alpha = \cos^{-1}(\frac{4}{9}) \approx 63{,}6°$ ... $\cos\alpha = \frac{4}{9}$

5. $\alpha = \tan^{-1}(\frac{14}{9}) \approx 57{,}3°$ ... $\tan\alpha = \frac{14}{9}$

6. $x = 2\sqrt{7289} \approx 170{,}8$ cm; $\alpha = \tan^{-1}(\frac{16}{170}) \approx 5{,}4°$ ...
$x^2 = 170^2 + 16^2$; $\tan\alpha = \frac{16}{170}$

7. $BC = \sqrt{20} \approx 4{,}47$; $\beta = \tan^{-1}(2) \approx 63{,}4°$ ... $BC^2 = 2^2 + 4^2$; $\tan\beta = \frac{4}{2}$

8. $a = \frac{7{,}4}{\sin 52°} \approx 9{,}4$ m ... $\sin 52° = \frac{7{,}4}{a}$

9. a) AB ≈ 1228,7 m ... $\sin 55° = AB : 1500$

b) AD ≈ 860,4 m ... $\cos 55° = AD : 1500$

10. $PB = \frac{1{,}20}{\tan 34{,}9°} \approx 1{,}72$ m ... $\tan 34{,}9° = 1{,}20 : PB$

11. $AC = \frac{h}{\sin 22°} \approx 78{,}1$ cm ... $h = \sqrt{32^2 - 13^2} = 3\sqrt{95}$; $\sin 22° = h : AC$

12. BF ≈ 93,0 cm ... $\sin 45° = BF : 131{,}5$
BC = BF : cos 65° ≈ 220,0 cm ... $\cos 65° = BF : BC$

13. Berg = 57,9 m; D = 69,9 m ... $\sin 30° = BC : 112{,}8$; Berg = BC + 1,5; D = 57,9 + 12,0

14. BD ≈ 502 m ... $\sin 54° = AD : 920$; $\tan 34° = BD : AD$

15. 100 und 6
ja ... $\frac{16}{170} = \frac{8}{85} \approx 9{,}4\ \% > 6\ \%$

16. $AC = 12{,}0 \cdot \frac{\sin 55°}{\sin 5°} \approx 112{,}8$ m ... $\gamma = 35° - 30°$; $AC : \sin 55° = \frac{12{,}0}{\sin 5°}$

17. $BD = 4{,}1 \cdot \frac{\sin 123°}{\sin 36°} \approx 5{,}85$ cm ... $BD : \sin 123° = \frac{4{,}1}{\sin 36°}$

18. $y = 160 \cdot \frac{\sin 141°}{\sin 34°} \approx 180{,}1$ cm ... $\frac{y}{\sin 141°} = \frac{160}{\sin 34°}$

19. $BC = 384 \cdot \frac{\sin 38°}{\sin 34°} \approx 422{,}8$ m ... $BC : \sin 38° = \frac{384}{\sin 34°}$

20. $BC = 4{,}50 \cdot \frac{\sin 62{,}8°}{\sin 34{,}9°} \approx 7{,}00$ m; A ≈ 92,0 m² ...
$BC : \sin 62{,}8° = \frac{4{,}50}{\sin 34{,}9°}$; $A = 8{,}00 \cdot (4{,}50 + 7{,}00)$

21. Weg ≈ 5665 m ... $JR : \sin 60° = \frac{2800}{\sin 50°}$; Weg = JR + 2500

22. nein ... $\angle DBC = 109° - 35° = 74°$; $BC : \sin 60° = \frac{2004}{\sin 74°}$; $BC = 2004 \cdot \frac{\sin 60°}{\sin 74°} \approx 1805{,}5$ m

23. $DP = 11 \cdot \frac{\sin 115°}{\sin 49°} - 10 \approx 3{,}2$ m ... $DE : \sin 115° = \frac{11}{\sin 49°}$; DP = DE − 10

24. $AD = 9 \cdot \frac{\sin 60°}{\sin 80°} \approx 7{,}9$ m ... $\angle ACD = 86° - 26° = 60°$; $AD : \sin 60° = \frac{9}{\sin 80°}$

25. AD, $\delta = 21°$ und rechter Winkel bei E
$A \approx 2{,}81$ cm² ... $AE = 4{,}1 \cdot \sin 21°$; $DE = 4{,}1 \cdot \cos 21°$